■ Stay-In School

AI-Powered Dropout Prevention System

RTGS AI Hackathon 2026 · Problem #100004

School Education Dept · Govt of Andhra Pradesh

34,512	**80.0%**	**9,149**	**26**
students flagged	dropout recall	schools covered	districts monit

The Problem

AP loses thousands of students every year — and we find out too late

1 in 12	65%	■0	6 months
students drops out by Class 10	dropouts are SC / ST / migrant children	intervention cost vs ■2L+ lost lifetime earnings	average delay between at-risk signal and action

■ Current system gaps:

- Attendance registers are checked monthly — absences compound silently
- Teachers have 40+ students per class — manual flagging is impossible
- No cross-dataset view: marks, income, migration, distance all siloed
- Dropout is confirmed only at year-end — intervention window is lost

Our Solution — Stay-In School

Four innovation pillars aligned to RTGS Problem Brief

AI-Powered Predictive Analytics

XGBoost trained on 4 govt datasets: attendance, FA/SA marks, GSWS socio-economic, migration flags. Flags at-risk students 3–6 months early.

Risk Visualization & Alerts

District heatmap, school roster with colour-coded tiers (Critical/High/Medium/Low). HM and teacher dashboards. SHAP driver sentences.

LEAP API Integration

Risk scores pushed to AP LEAP app via REST. Teacher interventions sync back — closing the retraining feedback loop.

Community & Parental Engagement

GenAI-drafted WhatsApp/SMS scripts in Telugu. Ward volunteer assignment. Scheme eligibility check (Amma Vodi, RTF/MTF, KGBV).

Meets RTGS success criteria: Exclusion error < 20% ✓ · Inclusion error < 80% ✓ · Explainable to non-technical staff ✓

Data Pipeline

4 government data streams → unified student risk profile

Dataset	Source	Key Features Extracted	Privacy
FIN_YEAR Attendance	School Ed. Dept	Attendance rate, consecutive absences, monthly trend	Role-scoped
FA / SA Marks	School Ed. Dept	Subject-wise scores, failure flags, score trend	Role-scoped
GSWS Household	GSWS Dept	Family income, parent literacy, migration flag	Aadhaar SHA-256 hashed
School Location Master	School Ed. Dept	Distance to school, rural/urban/tribal classification	Public data

Join key: CHILD_SNO (student ID) · Train: AY 2023-24 · Test (OOT): AY 2024-25 · 395,970 students · 9,149 schools

Model Performance

XGBoost · Out-of-time test on AY 2024-25 · Operating threshold p ≥ 0.5075

80.0%
Recall — dropouts caught early

12.0%
Precision — true positives among flagged

0.374
PR-AUC (precision-recall area)

0.942
ROC-AUC

Rank	Feature	SHAP Importance	Plain-English Label
1	migration_flag	26.6%	Migration / frequent relocation
2	caste_clean	11.2%	Social category (SC/ST/BC)
3	family_income_bracket	10.3%	Low family income
4	parent_literacy	10.1%	Low parent literacy
5	transport_allowance	6.2%	Long distance to school
6	n_present	5.8%	Low attendance days

Live Demo Flow

5-minute walkthrough — 4 screens, zero backend dependency

Step	Screen	What to show	Time
1	Home — /	4 innovation pillars, real metrics (recall/precision), demo credentials	30s
2	District Heatmap — /map	AP map with school risk dots; tap a school → details panel on right	45s
3	Model Overview — /overview	PR curve, fairness breakdown, feature importance bar chart	30s
4	Teacher Dashboard — /dashboard/teacher	Class roster colour-coded by tier; click a Critical student	45s
5	Student Detail	SHAP driver sentences in English + Telugu; counsellor card with parent SMS	60s
6	Community Card	WhatsApp button, ward volunteer name, ration card status, scheme alert	30s
7	AI Chatbot (floating)	"Vizag stats" → live district data; "boys vs girls" → gender recall breakdown	30s

SHAP Explainability

Every flag is auditable — no black box, no blind trust

Sample student explanation (SHAP TreeExplainer output):

Feature	Value	SHAP Contribution	Plain-English Sentence
migration_flag	1 (migrant)	**+34.2% ↑ risk**	This student's family has migrated recently — the strongest dropout signal.
attendance_rate	48%	**+18.7% ↑ risk**	Attendance below 50% — nearly half of school days missed this term.
family_income_bracket	Below poverty	**+12.1% ↑ risk**	Household income in lowest bracket — economic stress is a dropout driver.
transport_allowance	0 (no allowance)	**+8.4% ↑ risk**	School is >3km away with no transport support — barrier to attendance.
fa_avg	68 / 100	**–4.2% ↓ risk**	FA scores are reasonable — academic performance slightly reduces risk.

Teachers can contest any flag · Human-in-the-loop before any action is taken · Full audit trail logged

Community & Parental Engagement

GenAI-powered outreach in Telugu — reaching families where they are

What happens when a student is flagged Critical:

1. Class teacher gets a colour-coded alert in the dashboard.

2. A GenAI-drafted parent SMS is generated in Telugu:
"■■ ■■■■■■■■■■/■■■■■■ (■■■■) ■■■■■■■ ■■■■■■■■ ■■■■■■■■■■■■■■.
■■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■■■■■■■■■■■■■."

3. Ward volunteer is assigned (Asha Worker / GP Member / ANM).

4. WhatsApp direct-link button sends the message with one tap.

5. Intervention is logged → feeds retraining pipeline.

Schemes auto-suggested per student profile:

- **Amma Vodi** — ■15,000/yr for mothers (income check)
- **Vidya Kanuka** — free school kit (all enrolled)
- **Post-Matric RTF/MTF** — SC/ST/BC scholarship
- **NTR Vidyonnathi** — higher education aid
- **KGBV Residential** — girls in tribal/remote areas
- **Samagra Shiksha Transport** — students >3 km

Ethics & Data Privacy

DPDP Act 2023 compliant · Human-in-the-loop · Transparent AI

Principle	Implementation	Status
Aadhaar Protection	SHA-256 hash at ingest — raw number never enters the pipeline or browser	■ **Done**
Data Minimisation	Student names not stored in any browser-served JSON	■ **Done**
Role-scoped Access	Teachers see only their school · HMs see only their school cluster	■ **Done**
Human-in-the-loop	No automated action — all interventions require teacher confirmation	■ **Done**
Audit Trail	Every intervention logged with timestamp, method, volunteer ID	■ **Done**
Explainability	SHAP sentences in plain English + Telugu for every flag	■ **Done**
Infrastructure	Production: AP State Data Centre secure enclave, Amaravati	■ **Planned**

Scale Plan & Ask

3-district pilot → statewide in 12 months via LEAP API

Phase	Timeline	Scope	Key Milestone
Pilot	Month 1–3	3 districts · ~1,500 schools · ~1.2L students	Recall ≥ 80% validated on live data; teacher UX feedback
State Rollout	Month 4–9	All 26 districts · 9,149 schools · 3.96L students	LEAP API live integration; closed-loop retraining pipeline
Optimise	Month 10–12	Full AP + model v2 with intervention outcomes	Recall ≥ 85%; fairness gap < 5% across gender/caste

What we need from RTGS:

- Access to live LEAP API sandbox for integration testing
- 3-district pilot school data (2023-25)
- Coordination with School Ed. Dept GSWS team